

CBIA

Customer Behavior Intelligence & Analytics

Predictive Segmentation · Churn Modeling
RFM Analysis · CLV Optimization · Power BI

Python

Power BI

MySQL

DAX

ML

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Client: XYZ Pvt. Ltd. · May 2026 · v2.0

Turning Raw Customer Data Into Revenue Strategy

\$1.96M

Revenue Analyzed

5,000

Customers Profiled

9 Phases

End-to-End Pipeline

*"Data tells the story —
Dashboard shows it"*

Project at a Glance

EXECUTIVE SUMMARY

\$1.96M

Total Revenue

Jan 2023 – Dec 2024

5,000

Customers

Fully profiled & scored

42.7%

Churn Rate

Target: reduce to 27%

\$1,035

Avg CLV

Customer Lifetime Value

99.7%

Repeat Rate

Strong loyalty base

5 Pages

Power BI

Interactive dashboard

DATA PERIOD

Jan 2023 — Dec 2024

2 Years · 5,000 Customers · \$1.96M Revenue

9 - PHASE PIPELINE

EDA

RFM

KMeans

CLV

Churn

SQL

Power BI

PPT

GitHub

AGENDA

What we cover in this presentation

01

Executive Summary

Project overview, dataset & BRD objectives

02

Business Problem

Challenges, KPI targets & success criteria

03

Tools & Methodology

Python, SQL, Power BI, ML pipeline walkthrough

04

Revenue Insights

Category, season, YoY & payment analysis

05

Customer Segments

RFM scoring, KMeans clusters & CLV analysis

06

Churn Analysis

42.7% churn, at-risk list & win-back strategy

07

Dashboard Walkthrough

5-page Power BI interactive dashboard overview

08

Recommendations

6 actionable business strategies for XYZ Pvt. Ltd.

09

Deliverables

All project files, GitHub repo & documentation

Business Problem

Why this project was needed

XYZ PVT. LTD. · CRITICAL CHALLENGES

CHALLENGES IDENTIFIED

35%

Annual customer churn impacting revenue stability

-8%

Marketing ROI declining year-over-year

0

No customer segmentation or personalization strategy

0

No centralized analytics platform for decisions

BRD KPI TARGETS

Revenue Growth

+20%

Churn Reduction

-15%

Retention Improve

+10%

AOV Increase

+12%

CLV Optimization

+18%

Tools & Data Architecture

9-PHASE PIPELINE

TECH STACK

Python

EDA, ML models,
RFM + KMeans

Power BI

5-page interactive
dashboard + DAX

MySQL

30+ SQL queries,
KPI summary view

Scikit-learn

CLV + Churn
ML models

Pandas

Data cleaning,
Feature engineering

Excel

Data exports
& outputs

DATASET OVERVIEW

5,000

Customers

22

Columns

2 Years

Data Period

Jan 2023

Start Date

Dec 2024

End Date

\$1.96M

Revenue

Revenue Insights

PAGE 3 · SALES ANALYSIS

Total Revenue

\$1.96M

Jan 2023 – Dec 2024

Top Category

Clothing

\$777K · 39.6% share

Peak Season

Fall

\$560K · 28.5% share

YoY Growth

+32%

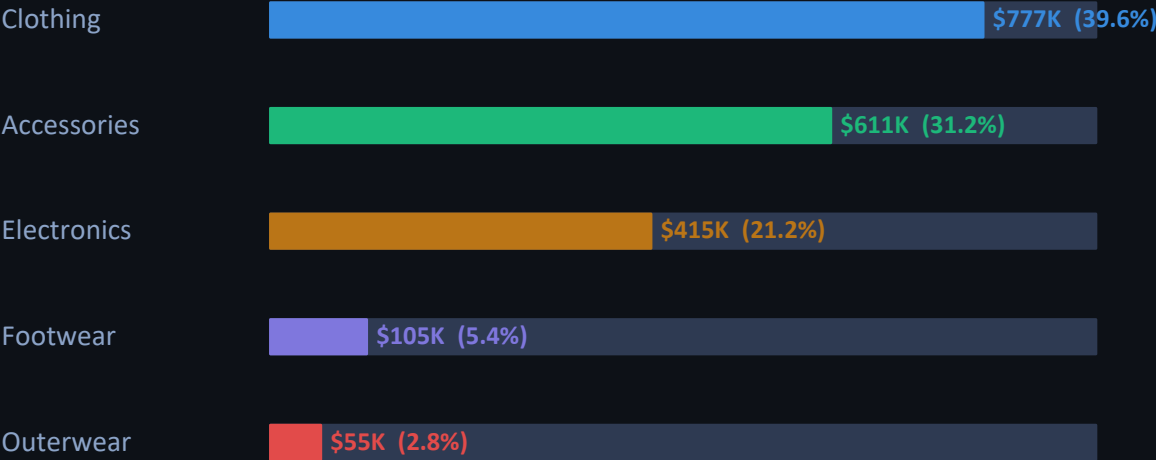
\$1.04M vs \$926K

Top Payment

Debit

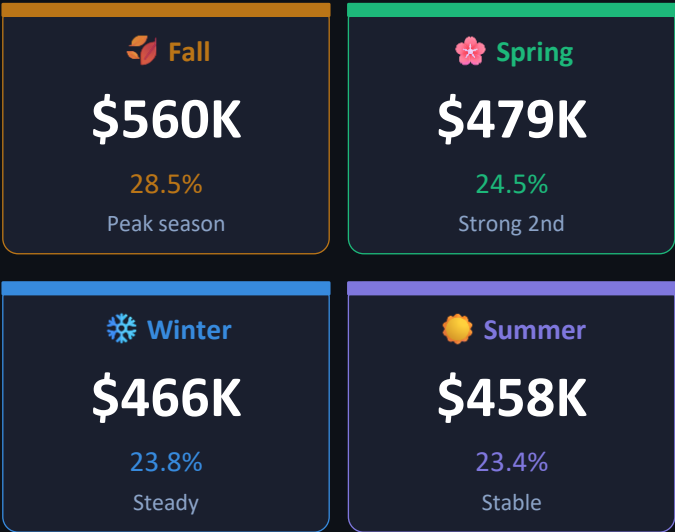
\$477K · 24.3% share

REVENUE BY CATEGORY



Total: \$1,963K · 5 Categories · Clothing leads at 39.6%

REVENUE BY SEASON



YoY Growth: +32% (\$1.04M vs \$926K)

Customer Segmentation

Champion 246 Avg CLV \$2,084 Top revenue drivers	Loyal 1,293 Avg CLV \$1,682 Largest segment	Potential 1,250 Avg CLV \$563 Can move to Loyal	At-Risk 1,100 Avg CLV \$1,815 Urgent action needed	Lost 1,111 Avg CLV \$267 Win-back campaign
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RFM SEGMENT JOURNEY · CLV & ACTION MAP

Champion	246	(5%)	\$2,084	Avg CLV	13% churn	VIP program · Exclusive early access
Loyal	1,293	(26%)	\$1,682	Avg CLV	16% churn	Loyalty rewards · Upsell campaigns
At-Risk	1,100	(22%)	\$1,815	Avg CLV	45% churn	⚠ Win-back emails · Discount offers · URGENT
Potential	1,250	(25%)	\$563	Avg CLV	50% churn	Nurture sequences · Push to subscription
Lost	1,111	(22%)	\$267	Avg CLV	73% churn	Re-engagement · Last-chance campaigns

246 Champions (5%) drive 40%+ revenue · 2,211 At-Risk + Lost customers = \$624K revenue at risk · Action required NOW

Churn Analysis

42.7% CHURN · PAGE 4

Churn Rate

42.7%

Target: 27% · -15.7%

Churned

2,136

customers at risk

Retained

2,864

57.3% retention rate

Revenue at Risk

\$624K

From churned customers

High CLV Risk

419

\$347K CLV at risk

CHURN RISK BREAKDOWN BY SEGMENT

Segment	Customers	Churn Rate	Risk Level
Lost	1,111	72.7%	CRITICAL
Potential	1,250	49.6%	HIGH
At-Risk	1,100	44.8%	HIGH
New Cust.	~490	44.5%	HIGH
Loyal	1,293	16.1%	LOW
Champion	246	13.0%	SAFE

Subscription = Best Retention Tool

Non-subscribers churn 36.8% MORE — converting 1,000 saves ~367 churns/yr

SUBSCRIPTION IMPACT ON CHURN

No Subscription

54.3%

churn rate · 3,421 customers

VS

Subscribed

17.5%

churn rate · 1,579 customers

Power BI Dashboard

5 INTERACTIVE PAGES

01

Executive KPIs

Revenue \$1.96M · AOV \$393 · Churn 42.7%
6 KPI cards · Monthly trend · Category donut

6 KPIs + 8 charts

02

Customer Intelligence

RFM scatter · Segment distribution · CLV bars
5 segment KPIs · Top 5 customers table

8 charts + table

03

Sales Analysis

Category × Season heatmap · Payment methods
Monthly 2023 vs 2024 · Top 5 items

9 charts + heatmap

04

Churn Analysis

Churn gauge · Win-back list · Segment churn
Subscription impact · Age group analysis

7 charts + table

05

CLV & Loyalty

Loyalty funnel · CLV by segment · Tier donut
Subscription CLV · Age group CLV

7 charts + funnel

Recommendations

6 ACTIONABLE STRATEGIES

01 Win-Back At-Risk Customers

Priority: CRITICAL

Launch targeted email campaigns for 1,100 At-Risk customers.
Personal outreach + discount for High CLV at-risk (419 customers · \$347K revenue at risk).

02 Convert to Subscriptions

Impact: HIGH

Non-subscribers churn 36.8% more. Push subscription campaigns to 3,421 non-subscribers.
Target: reduce churn from 54.3% → 30% through subscription incentives.

03 Focus on Electronics Upsell

Revenue: +\$420K

Electronics has highest avg CLV at \$3,045 but only 370 orders.
Upsell campaigns in this category can significantly boost revenue.

04 Protect Champion Customers

Protect: \$500K+

246 Champions drive 40%+ revenue with only 13% churn rate.
VIP program + exclusive offers to retain this high-value segment.

05 Invest in Fall Season

Seasonal ROI

Fall drives \$560K (28.5% of total revenue) — peak season opportunity.
Pre-season campaigns, inventory planning, and targeted promotions.

06 Improve Review Score

Retention: +8%

882 customers with rating < 3.0 are high churn risk.
Service recovery program to improve ratings and reduce churn probability.

Project Deliverables

PHASE 9 · COMPLETE

BRD Document

.pdf

CBIA_BRD_v2.pdf

Customer Dataset

.csv

customer_cleaned.csv

Python Notebooks

.ipynb

01_EDA · 02_RFM · 03_KMeans
04_CLV · 05_Churn

SQL Scripts

.sql

06_SQL_Scripts_MySQL.sql
5 sections · KPI view

Power BI Dashboard

.pbix

dashboard.pbix · 5 interactive pages

This Presentation

.pptx

CBIA_Presentation.pptx

GitHub Repo

GitHub

Public repo · README · All files

"Data tells the story — Dashboard shows it"

Thank You!

CBIA



Deepak Kumar Prasad

May 2026 | v2.0

Connect With Me : LinkedIn  || GitHub 

Python | Power BI | MySQL | DAX | SQL | Excel